College 2025-2026 Trimester 1

DETERMINING WINE QUALITY USING DECISION TREE, K-MEANS CLUSTERING AND LOGISTIC REGRESSION

EMC005 - DATAMINING

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TABLE OF CONTENTS

I. **PROBLEM STATEMENT** 3
I.I Project Objective *3*
II. **DATASET DESCRIPTION** 4
II.I Attributes Table *4*
III. **DATA EXPLORATION** 5
III.I Statistical Table *5*
IV. **DATA VISUALIZATION** 6
IV.I Histogram *6*
IV.II Bar Graph (Alcohol Content) *7*
IV.III Bar Graph (Volatile Acidity) *8*
IV.IV Heatmap *9*
V. **DATA PREPARATION** 10
V.I Data Cleaning *10*
V.II Data Preprocessing (Decision Tree and Logistic Regression) *10*
V.III Feature Selection (All Models) *10*
V.IV Data Normalization (k-Means and Logistic Regression) *10*
V.V For Clustering Technique (k-Mean) *11*
V.VI For Data Split Method (Decision Tree and Logistic Regression) *11*
V.VII For Cross Validation Method (Decision Tree and Logistic Regression) *11*
VI. **MODEL SELECTION** 12
VI.I Decision Tree Classification *12*
VI.II Clustering Technique (k-Mean) *12*
VI.III Logistic Regression *12*
VII. **RAPIDMINER PROCESS FLOW** 13
VII.I Dataset *13*
VII.II Decision Tree Classification *14*
VII.II.I Data Split Method 14
VII.II.II Cross Validation Method 15
VII.III Clustering Technique (k-Means) *16*
VII.IV Logistic Regression *17*
VII.IV.I Data Split Method 17
VII.IV.II Cross Validation Method 18
VIII. **EXPERIMENTS AND RESULTS** 19

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VIII.I Decision Tree Classification *19*
VIII.I.I Data Split Method 19
VIII.I.II Cross Validation Method 21
VIII.II Clustering Technique (k-Means) *23*
VIII.III Logistic Regression *25*
VIII.III.I Data Split Method 25
VIII.III.II Cross Validation Method 26
IX. **REFERENCE** 28

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I. Problem Statement

When it comes to the beverage manufacturing especially in wine, consistency of the quality of the products is essential in the success of the company, customer satisfaction and brand image. However, determining the wine quality by wine experts is a subjective measure that can be hard to predict. Although wine compositions have different measurements, there is no clear data-driven model we can use to determine the quality of wine produce based on these different physicochemical compositions (fixed acidity, volatile acidity, citric acid, etc.).

I.I Project Objective

This project objective is to address these problems with the use of data mining methods in determining the final product quality (score between 0 and 10) in relation to the physicochemical properties use during production.

This project will use **three (3)** different data-mining techniques:

Decision Tree Algorithm

- Identifying the quality classification of wine (Binary Classification: High or Standard).

Clustering Technique (k-Means)

- Identifying wine production pattern in determining natural grouping based on physicochemical compositions.

Logistic Regression

- For predicting the quality of wine based on wine compositions.

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II. Dataset Description

II.I Attributes Table

This data is from *UCI Machine Learning Repository* and using the *winequality-red.csv.*

Attribute Name	Feature Role	Data Type	Missing Values	Description
quality	Target	Integer	0	Sensory quality score (0-10)
fixed acidity	Predictor	Real	0	Primary fixed acids that influence taste
volatile acidity	Predictor	Real	0	Acetic acid content - high levels cause vinegar taste
citric acid	Predictor	Real	0	Adds freshness and flavor in small quantities
residual sugar	Predictor	Real	0	Sugar remaining after fermentation - affects sweetness
chlorides	Predictor	Real	0	Sodium chloride content - saltiness
free sulfur dioxide	Predictor	Real	0	Free SO2 - prevents microbial growth and oxidation
total sulfur dioxide	Predictor	Real	0	Total SO2 - antimicrobial and antioxidant effects
density	Predictor	Real	0	Density - influenced by alcohol and sugar content
pH	Predictor	Real	0	Acidity/basicity scale (0-14)
sulphates	Predictor	Real	0	Potassium sulphate - antioxidant and antimicrobial
alcohol	Predictor	Real	0	Alcohol content by volume (%)

Table 1: Attributes Table

The dataset has a total of:

- (**1,599**) examples
- (**12**) regular attributes
- (**0**) special attributes
- (**0**) no missing values

Target Attribute: *quality score (0-10 scale from sensory evaluation)*

Predictor: 11 physicochemical properties

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III. Data Exploration

III.I Statistical Table

Attribute Name	Data Type	Min	Max	Mean	Standard Deviation
quality	Integer	3	8	5.636	0.808
fixed acidity	Real	4.6	15.9	8.32	1.741
volatile acidity	Real	0.12	1.58	0.528	0.179
citric acid	Real	0	1	0.271	0.195
residual sugar	Real	0.9	15.5	2.539	1.41
chlorides	Real	0.012	0.611	0.087	0.047
free sulfur dioxide	Real	1	72	15.875	10.46
total sulfur dioxide	Real	6	289	46.468	32.895
density	Real	0.99	1.004	0.997	0.002
pH	Real	2.74	4.01	3.311	0.154
sulphates	Real	0.33	2	0.658	0.17
alcohol	Real	8.4	14.9	10.423	1.066

Table 2: Statistical Table

Key Findings:

- The quality distribution of wine is in median range (**5-6**).
- The minimum quality of wine is **3**, while the maximum quality is **8** (based on 0 – 10 scale)
- The **pH level** of wine average is within the acidity level **3.311**(acidity < pH 7, basicity > pH 7).
- There are instances that the wine that does not contains flavoring (**citric acid: min = 0**).
- Overall, alcohol has the most contributing factor, having a minimum content of **8.4** and a maximum of **14.9.** While chlorides have least content with average of **0.087**.

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IV. Data Visualization

IV.I Histogram

Table 3: Histogram - Wine Quality

Key Findings:

- The distribution is concentrated around standard quality wine (5.0 to 6.5).
- Majority of wine falls between 5.0 to 5.5 (**681 times**) and 6.0 to 6.5 (**638 times**).

This means that:

- In the dataset, majority of wine is in standard quality with **1382**.
- While wine with high quality only has **217.**
- The graph also shows a **slight right skewed**, because high-quality wine (7 to 8) is less frequent causing the tail extends to the right.

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IV.II Bar Graph (Alcohol Content)

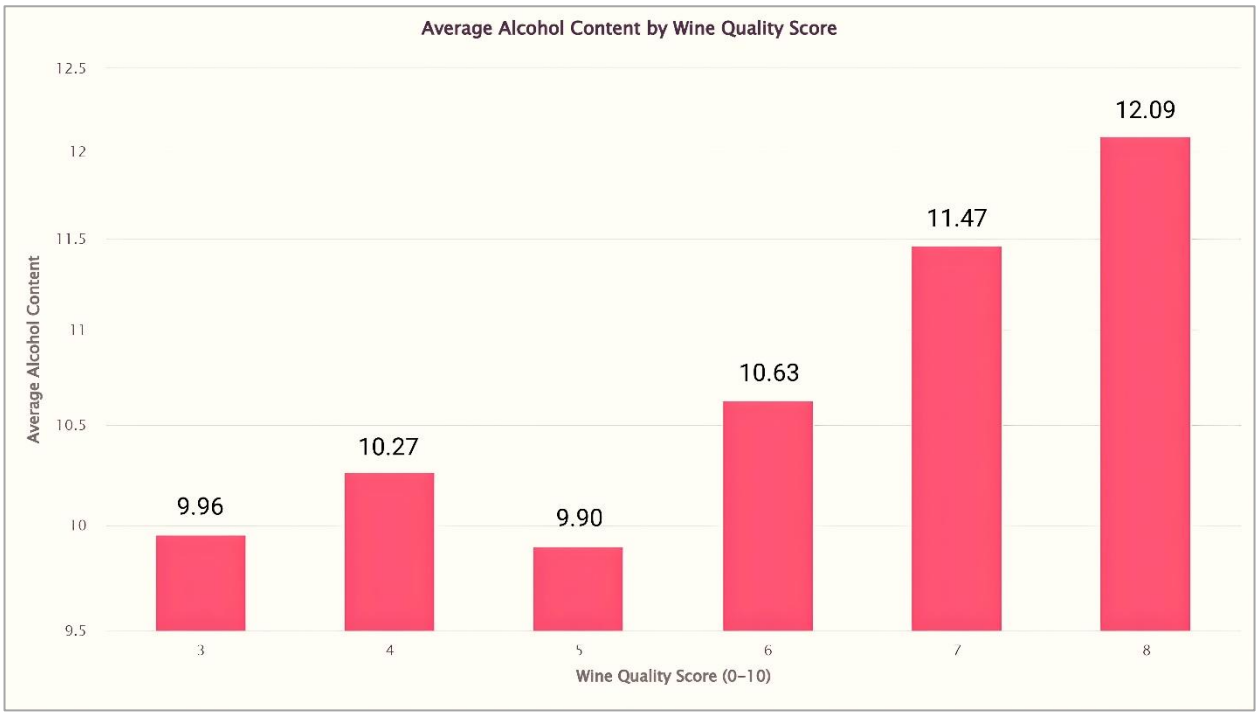

Table 4: Bar Graph - Alcohol Content

Key Findings:

- The graph shows that the highest average **(12.09%)** alcohol content belongs to wine with quality score of 8.
- While wine with quality score of 5 **(9.90%)** and 3 **(9.96%)** have the lowest alcohol content based on average.

This means that:

High quality wine **11.52%** (7 to 8) tends to have more alcohol contents than standard wine **10.25%** (3 to 6).

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IV.III Bar Graph (Volatile Acidity)

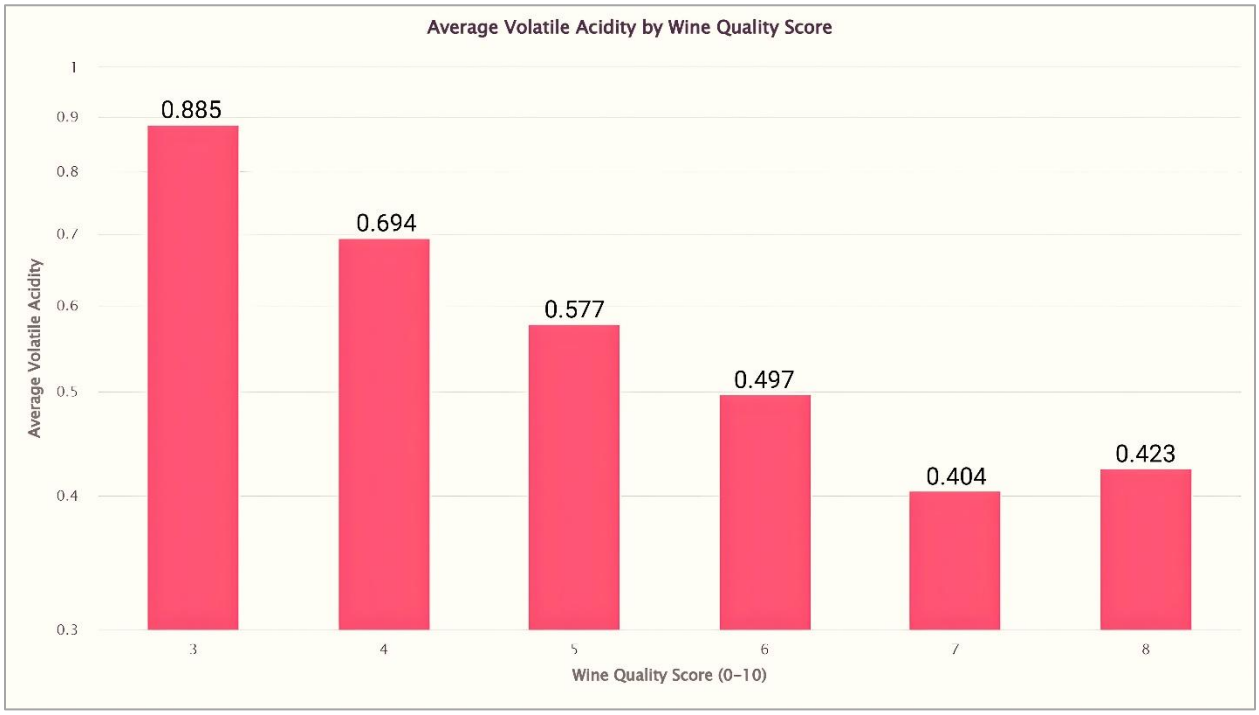

Table 5: Bar Graph - Volatile Acidity

Key Findings:

- The graph shows that the highest average **(0.885g/L)** volatile acidity belongs to wine with quality score of 3.
- While wine with quality score of 7 **(0.404g/L)** and 8 **(0.423g/L)** have low volatile acidity based on average.

This means that:

The more the wine quality increases (High: 7 to 8), the more it tends to have less volatile acidity compared to wine with lower quality (Standard: 3 to 6).

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IV.IV Heatmap

	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	free sulfur dioxide	total sulfur dioxide	density	pH	sulphates	alcohol	quality
fixed acidity	1	-0.256	0.672	0.115	0.094	-0.154	-0.113	0.668	-0.683	0.183	-0.062	0.124
volatile acidity	-0.256	1	-0.552	0.002	0.061	-0.011	0.076	0.022	0.235	-0.261	-0.202	-0.391
citric acid	0.672	-0.552	1	0.144	0.204	-0.061	0.036	0.365	-0.542	0.313	0.110	0.226
residual sugar	0.115	0.002	0.144	1	0.056	0.187	0.203	0.355	-0.086	0.006	0.042	0.014
chlorides	0.094	0.061	0.204	0.056	1	0.006	0.047	0.201	-0.265	0.371	-0.221	-0.129
free sulfur dioxide	-0.154	-0.011	-0.061	0.187	0.006	1	0.668	-0.022	0.070	0.052	-0.069	-0.051
total sulfur dioxide	-0.113	0.076	0.036	0.203	0.047	0.668	1	0.071	-0.066	0.043	-0.206	-0.185
density	0.668	0.022	0.365	0.355	0.201	-0.022	0.071	1	0.342	0.149	-0.496	-0.175
pH	-0.683	0.235	-0.542	-0.086	-0.265	0.070	-0.066	-0.342	1	-0.197	0.206	-0.058
sulphates	0.183	-0.261	0.313	0.006	0.371	0.052	0.043	0.149	-0.197	1	0.094	0.251
alcohol	-0.062	-0.202	0.110	0.042	-0.221	-0.069	-0.206	-0.496	0.206	0.094	1	0.476
quality	0.124	-0.391	0.226	0.014	-0.129	-0.051	-0.185	-0.175	-0.058	0.251	0.476	1

−1 −0.5 0 0.5 1

Table 6: Correlation Matrix - Heatmap

Key Findings:

- The **top 5 factors** associated with our target attribute quality are *alcohol, sulphates, citric acid, density, and volatile acidity.* These 5 attributes have the highest correlation to *quality* (close to 1 or -1)
- *alcohol* has the **highest positive correlation (47.6%)** with our target attribute *quality,* while *volatile acidity* has the **highest negative correlation (-39.1%)** with our target attribute *quality.*

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V. Data Preparation

This section shows essential steps in preparation for model training and evaluation.

V.I Data Cleaning

The dataset is well maintained with zero (0) missing values and no duplicates.

V.II Data Preprocessing (Decision Tree and Logistic Regression)

Converting the original target attribute *quality (0 to 10)* into binary classification:

- **High Quality**: *quality* ≥ 7
- **Standard Quality**: *quality* ≤ 6

*New generated attribute: **binary_quality** (standard and high).

This will help in reducing model bias because of the class imbalance shown in *Table 3: Histogram – Wine Quality.*

V.III Feature Selection (All Models)

Using *Table 6: Correlation Matrix – Heatmap,* these are the top 5 features selected that have strong relationship to our target attribute.

- *alcohol* (+0.476)
- *sulphates* (+0.251)
- *citric acid* (+0.226)
- *density* (-0.175)
- *volatile acidity* (-0.391)

This can help in reducing noise that came from the low correlation predictors.

V.IV Data Normalization (k-Means and Logistic Regression)

This will be used in k-Means and logistic regression to balance the featured selected attributes to prevent biases. This will convert the features attributes into standard deviation with the mean of zero (0).

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V.V For Clustering Technique (k-Mean)

- Value for k: 2

This separates the wine quality into two distinct groups (standard and high).

V.VI For Data Split Method (Decision Tree and Logistic Regression)

The dataset will be split into:

- Training Set: 60%
- Validation Set: 20%
- Testing Set: 20%

60% for training the data, 20% validation to avoid overfitting and 20% testing unbiased data.

V.VII For Cross Validation Method (Decision Tree and Logistic Regression)

- Number of folds: 10
- 90% Training
- 10% Testing

This will train the model 10 times and get the average results from the 10% testing scores.

This method will be used to compare the results when using data split method.

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VI. Model Selection

These three (3) models were selected to solve the problem in quality using classification (Decision Tree), prediction (Logistic Regression), and unsupervised learning (k-Means Clustering).

VI.I Decision Tree Classification

- This model will identify the wine quality classification whether it is **standard** or **high** quality.

This can be used for quality control in manufacturing.

VI.II Clustering Technique (k-Mean)

- This model will identify wine production pattern to determine natural grouping based on the physicochemical compositions.

This can be used to identify wine groupings based on different styles.

VI.III Logistic Regression

- This model will predict the wine quality probability based on featured selected wine compositions.

This can be used as statistical analysis on what specific physicochemical compositions affect the quality of wine during manufacturing.

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VII. RapidMiner Process Flow

VII.I Dataset

Row No.	fixed acidity	volatile acid...	citric acid	residual sug...	chlorides	free sulfur d...	total sulfur ...	density	pH	sulphates	alcohol	quality
1	7.400	0.700	0	1.900	0.076	11	34	0.998	3.510	0.560	9.400	5
2	7.800	0.880	0	2.600	0.098	25	67	0.997	3.200	0.680	9.800	5
3	7.800	0.760	0.040	2.300	0.092	15	54	0.997	3.260	0.650	9.800	5
4	11.200	0.280	0.560	1.900	0.075	17	60	0.998	3.160	0.580	9.800	6
5	7.400	0.700	0	1.900	0.076	11	34	0.998	3.510	0.560	9.400	5
6	7.400	0.660	0	1.800	0.075	13	40	0.998	3.510	0.560	9.400	5
7	7.900	0.600	0.060	1.600	0.069	15	59	0.996	3.300	0.460	9.400	5
8	7.300	0.650	0	1.200	0.065	15	21	0.995	3.390	0.470	10	7
9	7.800	0.580	0.020	2	0.073	9	18	0.997	3.360	0.570	9.500	7
10	7.500	0.500	0.360	6.100	0.071	17	102	0.998	3.350	0.800	10.500	5
11	6.700	0.580	0.080	1.800	0.097	15	65	0.996	3.280	0.540	9.200	5
12	7.500	0.500	0.360	6.100	0.071	17	102	0.998	3.350	0.800	10.500	5
13	5.600	0.615	0	1.600	0.089	16	59	0.994	3.580	0.520	9.900	5
14	7.800	0.610	0.290	1.600	0.114	9	29	0.997	3.260	1.560	9.100	5
15	8.900	0.620	0.180	3.800	0.176	52	145	0.999	3.160	0.880	9.200	5
16	8.900	0.620	0.190	3.900	0.170	51	148	0.999	3.170	0.930	9.200	5
17	8.500	0.280	0.560	1.800	0.092	35	103	0.997	3.300	0.750	10.500	7
18	8.100	0.560	0.280	1.700	0.368	16	56	0.997	3.110	1.280	9.300	5
19	7.400	0.590	0.080	4.400	0.086	6	29	0.997	3.380	0.500	9	4
20	7.900	0.320	0.510	1.800	0.341	17	56	0.997	3.040	1.080	9.200	6
21	8.900	0.220	0.480	1.800	0.077	29	60	0.997	3.390	0.530	9.400	6
22	7.600	0.390	0.310	2.300	0.082	23	71	0.998	3.520	0.650	9.700	5
23	7.900	0.430	0.210	1.600	0.106	10	37	0.997	3.170	0.910	9.500	5
24	8.500	0.490	0.110	2.300	0.084	9	67	0.997	3.170	0.530	9.400	5
25	6.900	0.400	0.140	2.400	0.085	21	40	0.997	3.430	0.630	9.700	6

ExampleSet (1,599 examples,0 special attributes,12 regular attributes)

Table 7: winequality - red

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VII.II Decision Tree Classification

VII.II.I Data Split Method

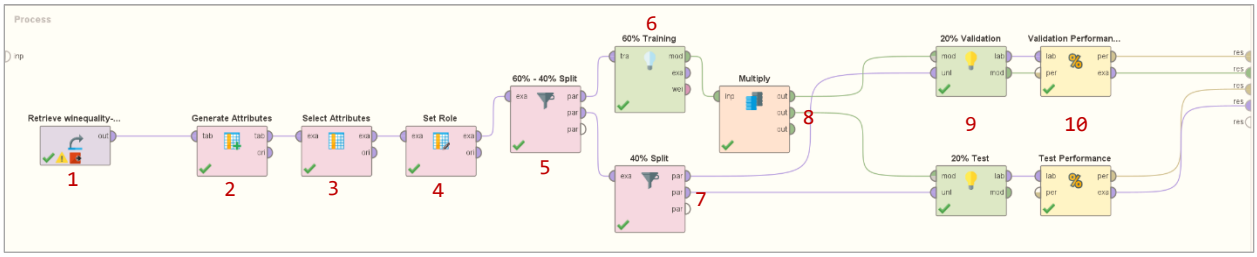

Table 8.1: Split Data Process – Decision Tree Classification

Process Flow

1. **Load the dataset** *winequality-red.csv.*
2. **Generate Attribute**: Create binary_quality using expression: if(quality>=7,"high","standard")
3. **Select Attribute**: use Top 5 Feature Selection attributes and include *binary_quality.*
4. **Set Role:** Set *binary_quality* attribute into label as the target attribute.
5. **1st Split Data**: Use ratio 0.6 and 0.4
6. **Decision Tree Operator:** Use first partition 0.6 as training set.
7. **2nd Split Data:** Use ratio 0.2 for validation set and test set.
8. **Multiply:** Use in training set to connect in validation set and test set.
9. **Apply Model:** Connect the multiply from training set into both model of validation set and test set. Also connect the two partitions from step 7 into the unlabeled data port in validation set and test set.
10. **Performance Operator:** Check the accuracy of validation and test outputs.

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VII.II.II Cross Validation Method

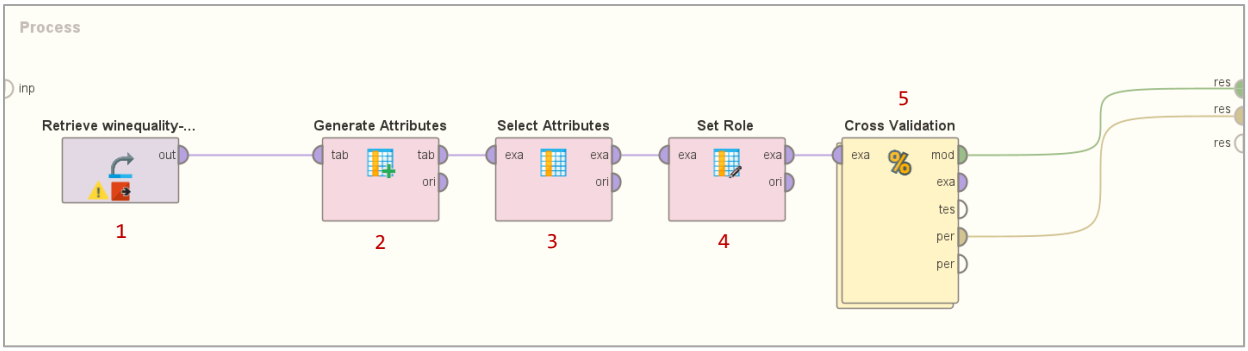

Table 8.2: Cross Validation Process - Decision Tree Classification

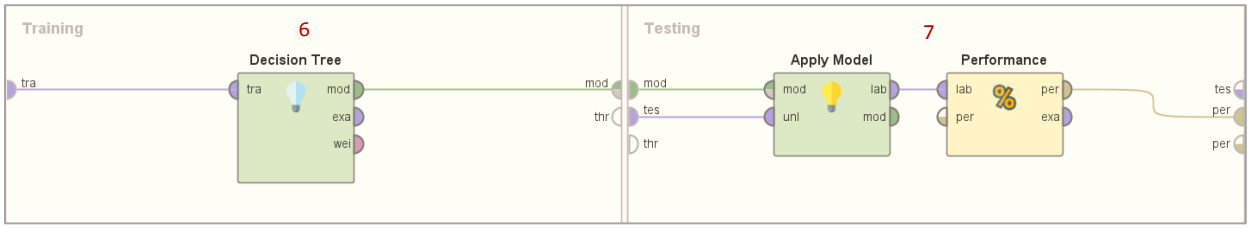

Table 8.3: Cross Validation Process - Decision Tree Classification

Process Flow

1. **Load the dataset** *winequality-red.csv.*
2. **Generate Attribute**: Create binary_quality using expression:

   if(quality>=7,"high","standard")
3. **Select Attribute**: use Top 5 Feature Selection attributes and include *binary_quality.*
4. **Set Role:** Set *binary_quality* attribute into label as the target attribute.
5. **Cross Validation:** Use 10 folds and leave sampling type to automatic.
6. **Cross Validation (Training)**: Insert Decision Tree Operator in training column.
7. **Cross Validation (Testing)**: Insert Apply Model and Performance in training column.

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VII.III Clustering Technique (k-Means)

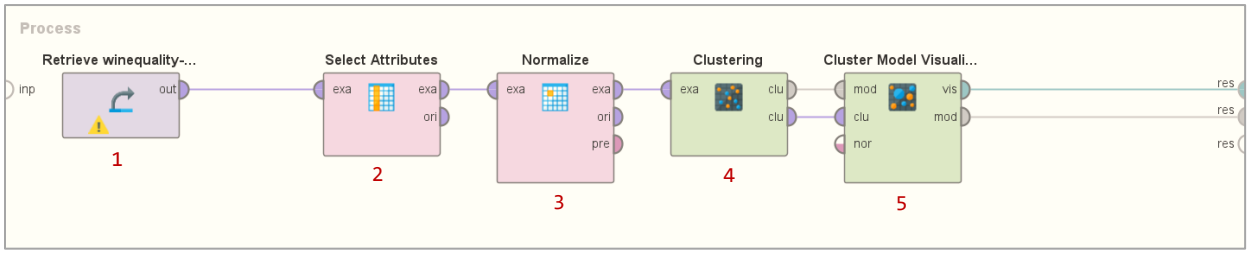

Table 9: Clustering Technique Process

Process Flow

1. **Load the dataset** *winequality-red.csv.*
2. **Select Attribute**: Use Top 5 Feature Selection attributes.
3. **Normalize:** Normalize all attributes and use Z-transform.
4. **Clustering:** Insert Clustering Operator
5. **Cluster Model Visualizer:** Optional operator to see clustering model in different graphs.

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VII.IV Logistic Regression

VII.IV.I Data Split Method

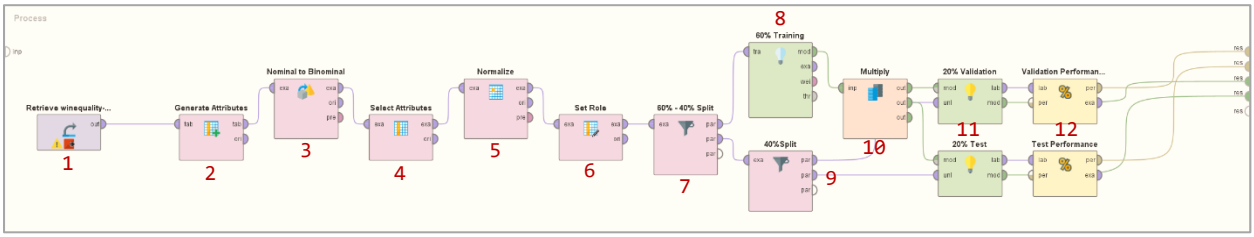

Table 10.1: Split Data Process - Logistic Regression

Process Flow

1. **Load the dataset** *winequality-red.csv.*
2. **Generate Attribute**: Create binary_quality using expression: if(quality>=7,"high","standard")
3. **Nominal to Binominal**: Convert binary_quality attribute into binominal.
4. **Select Attribute**: use Top 5 Feature Selection attributes and include *binary_quality.*
5. **Normalize:** Normalize all attributes and use Z-transform.
6. **Set Role:** Set *binary_quality* attribute into label as the target attribute.
7. **1st Split Data**: Use ratio 0.6 and 0.4
8. **Logistic Regression Operator:** Use first partition 0.6 as training set.
9. **2nd Split Data:** Use ratio 0.2 for validation set and test set.
10. **Multiply:** Use in training set to connect in validation set and test set.
11. **Apply Model:** Connect the multiply from training set into both model of validation set and test set. Also connect the two partitions from step 7 into the unlabeled data port in validation set and test set.
12. **Performance Operator:** Check the accuracy of validation and test outputs.

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VII.IV.II Cross Validation Method

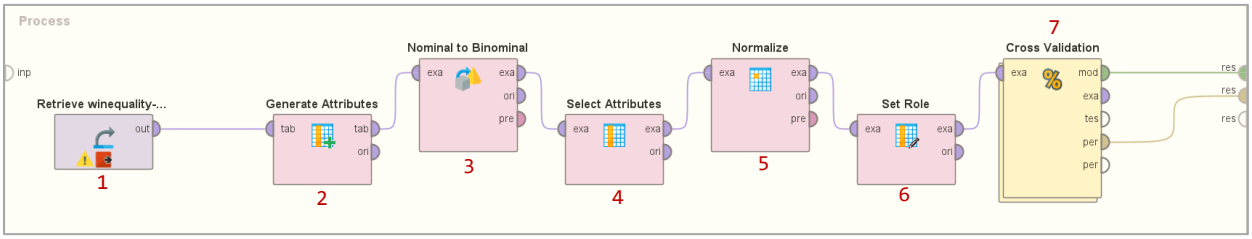

Table 10.2: Cross Validation Process - Logistic Regression

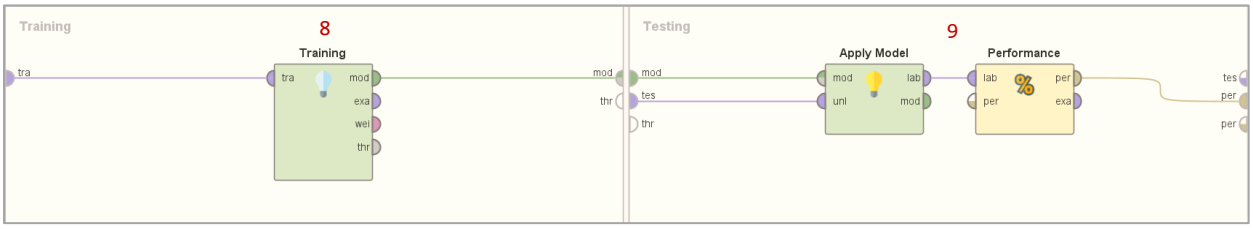

Table 10.3: Cross Validation Process - Logistic Regression

Process Flow

1. **Load the dataset** *winequality-red.csv.*
2. **Generate Attribute**: Create binary_quality using expression:

   if(quality>=7,"high","standard")
3. **Nominal to Binominal**: Convert binary_quality attribute into binominal.
4. **Select Attribute**: use Top 5 Feature Selection attributes and include *binary_quality.*
5. **Normalize:** Normalize all attributes and use Z-transform.
6. **Set Role:** Set *binary_quality* attribute into label as the target attribute.
8. **Cross Validation:** Use 10 folds and leave sampling type to automatic.
7. **Cross Validation (Training)**: Insert Logistic Regression Operator in training column.
8. **Cross Validation (Testing)**: Insert Apply Model and Performance in training column.

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VIII. Experiments and Results

VIII.I Decision Tree Classification

VIII.I.I Data Split Method

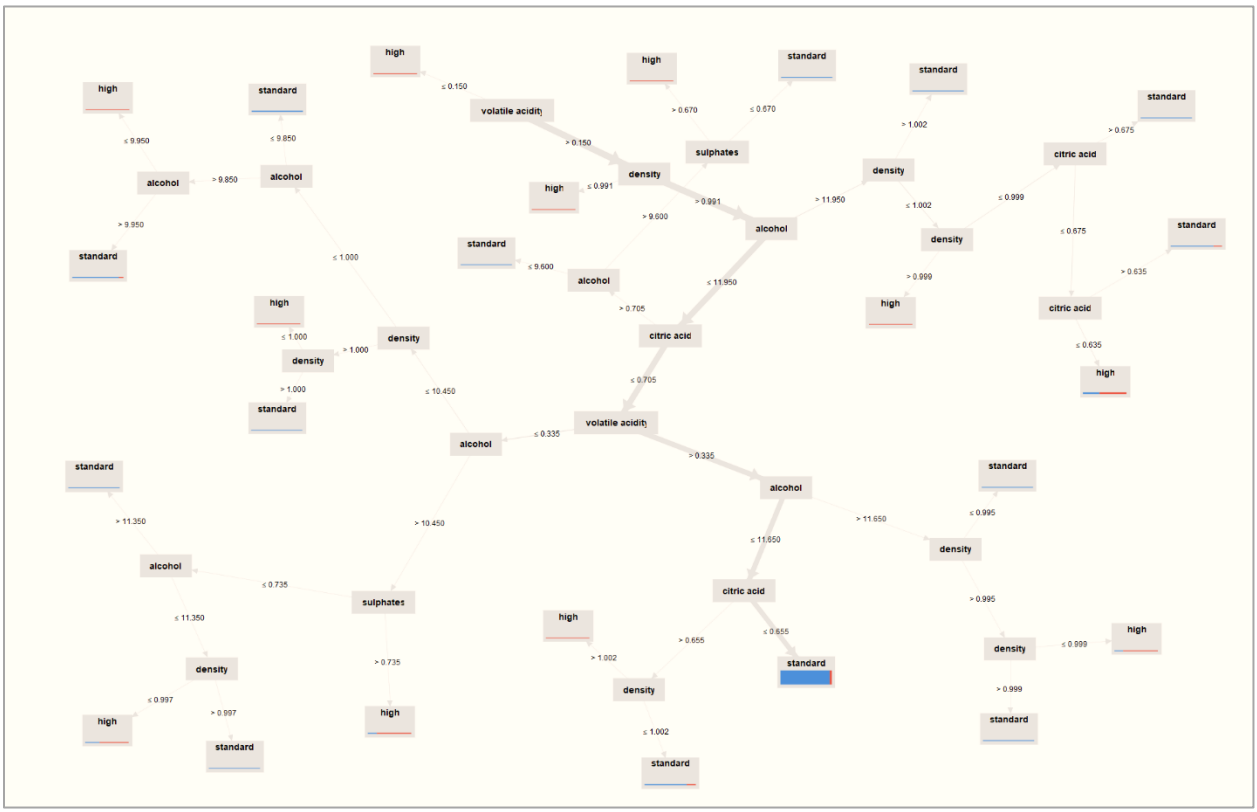

Table 11.1: ISOM View – Decision Tree Classification

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accuracy: 75.62%

	true standard	true high	class precision
pred. standard	203	29	87.50%
pred. high	49	39	44.32%
class recall	80.56%	57.35%	

Table 11.2: Validation Accuracy

Test Performance Vector

- **accuracy:** 75.62%
- **precision:** 44.32% (positive class: high)
- **recall:** 57.35% (positive class: high)
- **AUC (Area Under Curve):** 0.670 (positive class: high)

accuracy: 83.44%

	true standard	true high	class precision
pred. standard	258	16	94.16%
pred. high	37	9	19.57%
class recall	87.46%	36.00%	

Table 11.3: Test Accuracy

Test Performance Vector

- **accuracy:** 83.44%
- **precision:** 19.57% (positive class: high)
- **recall:** 36.00% (positive class: high)
- **AUC (Area Under Curve):** 0.634 (positive class: high)

The Decision Tree model using split data (60-20-20) achieved **83.44% overall accuracy** in test set. In a normal setting, this means that for every **100** wines batches, **83** wines are correctly identified as standard or high quality by only checking the physicochemical compositions.

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VIII.I.II Cross Validation Method

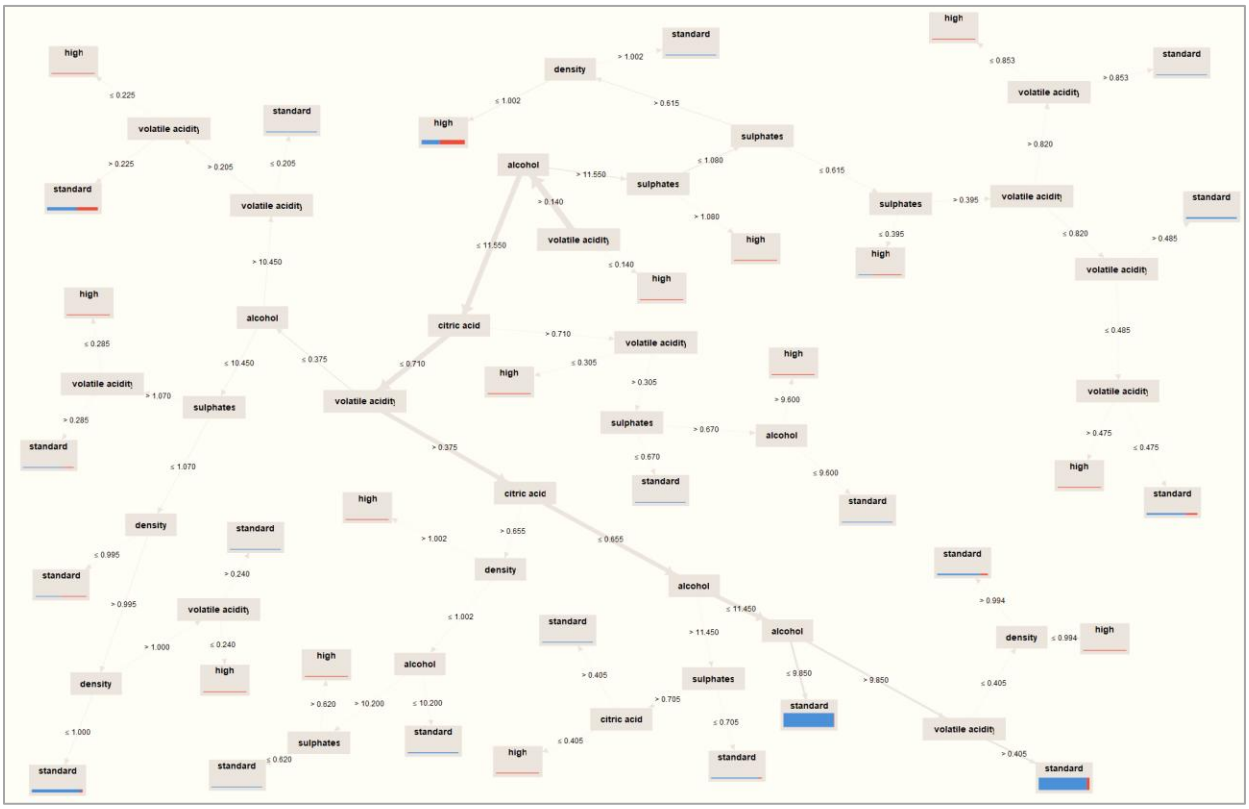

Table 12.1: ISOM View – Decision Tree Classification

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accuracy: 86.81% +/- 1.59% (micro average: 86.80%)

	true standard	true high	class precision
pred. standard	1319	148	89.91%
pred. high	63	69	52.27%
class recall	95.44%	31.80%	

Table 12.2: Test Accuracy

Test Performance Vector

- **accuracy:** 86.81% +/- 1.59% (micro average: 86.80%)
- **precision:** 54.46% +/- 13.33% (micro average: 52.27%) (positive class: high).
- **recall:** 31.82% +/- 14.33% (micro average: 31.80%) (positive class: high).
- **AUC (Area Under Curve):** 0.787 +/- 0.037 (micro average: 0.787) (positive class: high)

The Decision Tree model using cross validation achieved **86.80% overall accuracy** in test set. This means that for every **100** wines batches, **87** wines are correctly identified as standard or high quality by only checking the physicochemical compositions.

Performance Vector	***Split Data Method***	***Cross Validation Method***
accuracy	**83.44%**	**86.80%**
precision	**19.57%**	**52.27%**
recall	**36.00%**	**31.80%**
AUC	**63.4%**	**78.7%**

Table 12.3: Comparison Table

Conclusion:

Decision Tree model using cross validation method is the best way to use in classifying wine whether it is standard or high quality because of its significant increase in precision. This model can be useful for business decision making primarily in quality control and quality assurance.

This can also help business operations to be efficient in classifying standard and high-quality wines.

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VIII.II Clustering Technique (k-Means)

Cluster Model

- Cluster 0: 893 items – Standard Quality
- Cluster 1: 706 items – High Quality

Total number of items: 1599

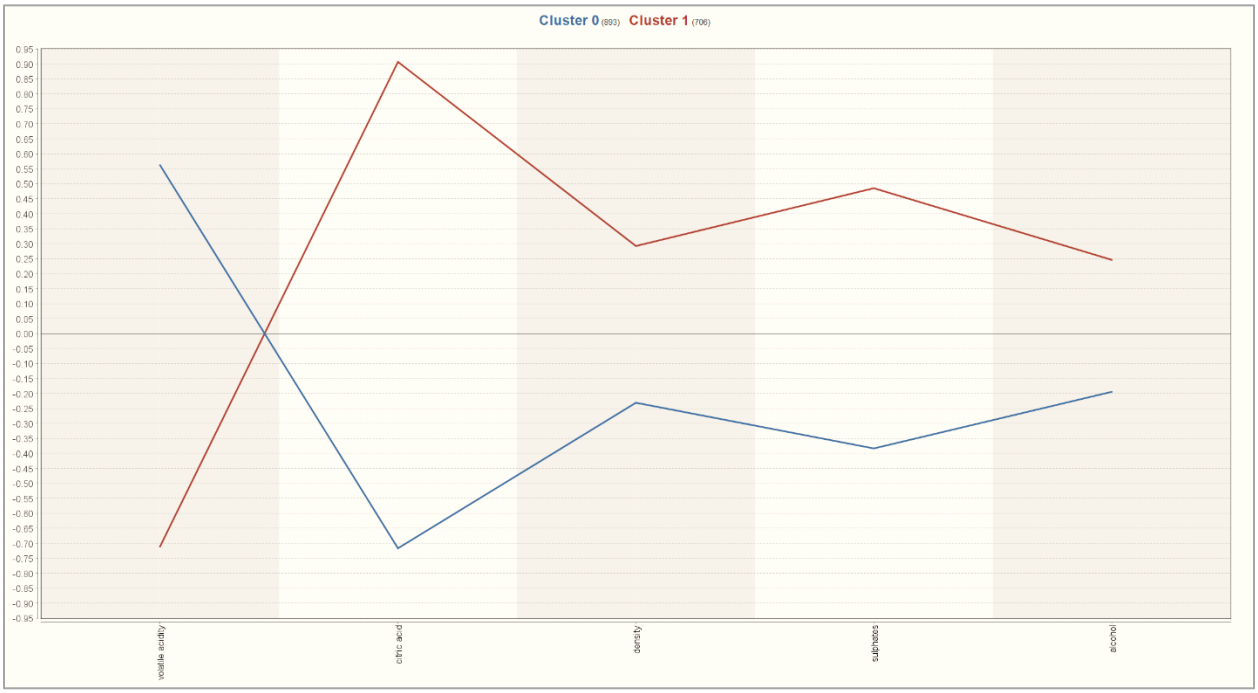

Table 13.1: Centroid Table - Graph

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Cluster	volatile acidity	citric acid	density	sulphates	alcohol
Cluster 0	0.562	-0.717	-0.231	-0.383	-0.194
Cluster 1	-0.711	0.907	0.292	0.485	0.246

Table 13.1: Centroid Table

The results shows that cluster 0 was considered as *standard quality* wine based on the 5 featured selected physicochemical compositions:

- Volatile acidity content is high (0.562) meaning the wine has more acetic acid content (vinegar taste).
- Citric acid content is very low (-0.717), this means that the wine has less freshness and flavoring.
- The other three (3) physicochemical compositions also showing less content when the average is 0 (by using normalization).

While cluster 1 was considered as *high quality* wine because:

- Volatile acidity content is very low (-0.711) meaning the wine is more pure wine quality and has less acetic acid content.
- Citric acid content is very high (0.907), this means that the wine was more balance between freshness and flavoring.
- The other three (3) physicochemical compositions also showing more content when the average is 0 (by using normalization).

Conclusion:

The k-Mean clustering model has effectively grouped the wines into two distinct category (standard and high-quality) wine just by analyzing their physicochemical compositions. The two physicochemical compositions that have major impact in the quality of wine were the volatile acidity and citric acid.

In business, this can be helpful in segregating the wine easily just by checking the volatile content and citric acid contents.

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VIII.III Logistic Regression

VIII.III.I Data Split Method

Attribute	Coefficient	Std. Coefficient	Std. Error	z-Value	p-Value
volatile acidity	-0.505	-0.486	0.164	-3.081	0.002
citric acid	0.212	0.213	0.160	1.325	0.185
density	-0.025	-0.025	0.139	-0.183	0.855
sulphates	0.552	0.500	0.112	4.922	0.000 = 8.546E-7
alcohol	0.918	0.941	0.128	7.143	0.000 = 9.156E-13
Intercept	-2.545	-2.534	0.152	-16.707	0

Table 14.1: Model Table

Checking at the model coefficient table, volatile acidity (***β = -0.505, ρ < 0.002***), sulphates (***β = 0.552, ρ < 0.001***), alcohol (***β = 0.918, ρ < 0.001***) were statistically significant predictors. While citric acid and density were not statistically significant predictors.

accuracy: 86.25%

	true standard	true high	class precision
pred. standard	243	11	95.67%
pred. high	33	33	50.00%
class recall	88.04%	75.00%	

Table 14.2: Validation Accuracy

Test Performance Vector

- **accuracy:** 86.25%
- **precision:** 50.00% (positive class: high)
- **recall:** 75.00% (positive class: high)
- **AUC (Area Under Curve):** 0.919 (positive class: high)

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accuracy: 82.50%

	true standard	true high	class precision
pred. standard	234	13	94.74%
pred. high	43	30	41.10%
class recall	84.48%	69.77%	

Table 14.3: Test Accuracy

Test Performance Vector

- **accuracy:** 82.50%
- **precision:** 41.10% (positive class: high)
- **recall:** 69.77% (positive class: high)
- **AUC (Area Under Curve):** 0.848 (positive class: high)

The Logistic Regression model using split data (60-20-20) achieved **82.50% overall accuracy** in test set. In a normal setting, this means that for every **100** wines batches, **83** wines are correctly identified as standard or high quality by analyzing volatile acidity, alcohol content and sulphates.

VIII.III.II Cross Validation Method

Attribute	Coefficient	Std. Coefficient	Std. Error	z-Value	p-Value
volatile acidity	-0.671	-0.671	0.132	-5.081	0.000 = 3.750E-7
citric acid	0.135	0.135	0.128	1.055	0.291
density	0.070	0.070	0.112	0.630	0.529
sulphates	0.411	0.411	0.077	5.364	0.000 = 8.146E-8
alcohol	1.117	1.117	0.108	10.377	0 = 3.156E-25
Intercept	-2.652	-2.652	0.125	-21.145	0

Table 15.1: Model Table

Checking at the model coefficient table, volatile acidity (***β = -0.671, ρ < 0.001***), sulphates (***β = 0.411, ρ < 0.001***), alcohol (***β = 1.117, ρ < 0.001***) were statistically significant predictors. While citric acid and density were not statistically significant predictors.

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accuracy: 84.24% +/- 2.69% (micro average: 84.24%)

	true standard	true high	class precision
pred. standard	1200	70	94.49%
pred. high	182	147	44.68%
class recall	86.83%	67.74%	

Table 15.2: Test Accuracy

Test Performance Vector

- **accuracy:** 84.24% +/- 2.69% (micro average: 84.24%)
- **precision:** 44.92% +/- 6.46% (micro average: 44.68%) (positive class: high)
- **recall:** 67.81% +/- 10.10% (micro average: 67.74%) (positive class: high)
- **AUC (Area Under Curve):** 0.862 +/- 0.029 (micro average: 0.862) (positive class: high)

The Logistic Regression model using cross validation achieved **84.24% overall accuracy** in test set. This means that for every **100** wines batches, **84** wines are correctly identified as standard or high quality by analyzing volatile acidity, alcohol content and sulphates.

Performance Vector	***Split Data Method***	***Cross Validation Method***
accuracy	**82.50%**	**84.24%**
precision	**41.10%**	**44.68%**
recall	**69.77%**	**67.74%**
AUC	**84.8%**	**86.2%**

Table 15.3: Comparison Table

Conclusion:

Based on the results, there is no significant difference in the performance between the two methods. But cross validation method is recommended because 3 out of 4 performances had increased.

In business, this model can reduce the cost of having professional wine taster. You will just use this model to predict the wine quality by checking its volatile acidity, sulphates and alcohol contents.

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[Pre-press (pdf)] http://www3.dsi.uminho.pt/pcortez/winequality09.pdf

[bib] http://www3.dsi.uminho.pt/pcortez/dss09.bib

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